ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **World of Algorithms** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 549	Add column by column from the right, carrying 1 when a column sum exceeds 9.
2	**(b)** 8	Divisors of 24: 1, 2, 3, 4, 6, 8, 12, 24.
3	**(d)** 2	$38 = 12 \times 3 + 2$.
4	**(b)** 2	Common divisors: 1, 2.
5	**(d)** a step-by-step procedure that solves a problem	Definition.
6	**(b)** 30	Sum of even numbers up to 11.
7	**(d)** gcd(n, m − n) for $m \geq n$	Common divisors divide the difference.
8	**(d)** 1	Carries out of the columns: [0, 1].
9	**(a)** 561	Add column by column from the right, carrying 1 when a column sum exceeds 9.
10	**(b)** 6	Divisors of 12: 1, 2, 3, 4, 6, 12.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Add from the units column, carry 1 when a column sum exceeds 9; Columns with a carry: units (1); $318 + 427 = 745$ (1).	1+1
12	Check each j from 1 to 14: 14 numbers are checked; List $= [1, 2, 7, 14]$ (1); It has 4 entries (1).	1+1
13	$81 = 12 \times 6 + 9$, so remainder 9; $93 = 12 \times 7 + 9$, so remainder 9 (1); Adding the divisor does not change the remainder (1).	1+1
14	Add from the units column, carry 1 when a column sum exceeds 9; Columns with a carry: none (1); $246 + 330 = 576$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	divisors(10) $= [1, 2, 5, 10]$ (1); divisors(30) $= [1, 2, 3, 5, 6, 10, 15, 30]$ (1); Common divisors $= [1, 2, 5, 10]$, greatest is 10 (1).	1+1+1
16	Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first (1); $(39, 27) \to (27, 12) \to (15, 12) \to (12, 3) \to (9, 3) \to (6, 3) \to (3, 3) \to (3, 0)$ (1); 7 subtractions, gcd $= 3$ (1).	1+1+1
17	Add from the units column, carry 1 when a column sum exceeds 9 (1); Columns with a carry: hundreds, thousands (1); $5840 + 4426 = 10266$ (1).	1+1+1
	OR: divisors(33) $= [1, 3, 11, 33]$ (1); divisors(18) $= [1, 2, 3, 6, 9, 18]$ (1); Common divisors $= [1, 3]$, greatest is 3 (1).	1+1+1
18	Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first (1); $(42, 28) \to (28, 14) \to (14, 14) \to (14, 0)$ (1); 3 subtractions, gcd $= 14$ (1).	1+1+1
	OR: Add from the units column, carry 1 when a column sum exceeds 9 (1); Columns with a carry: units (1); $3057 + 3038 = 6095$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Even values: 2, 4, 6, 8, 10, 12, 14 (1); Total $= 2 + 4 + 6 + 8 + 10 + 12 + 14$ (2); Final total $= 56$ (2).	1+2+2
	OR: divisors(50) $= [1, 2, 5, 10, 25, 50]$ (1); divisors(75) $= [1, 3, 5, 15, 25, 75]$ (2); Common divisors $= [1, 5, 25]$, greatest is 25 (2).	1+2+2
20	Even values: 2, 4, 6, 8, 10 (1); Total $= 2 + 4 + 6 + 8 + 10$ (2); Final total $= 30$ (2).	1+2+2
	OR: divisors(66) $= [1, 2, 3, 6, 11, 22, 33, 66]$ (1); divisors(84) $= [1, 2, 3, 4, 6, 7, 12, 14, 21, 28, 42, 84]$ (2); Common divisors $= [1, 2, 3, 6]$, greatest is 6 (2).	1+2+2

Mistakes to avoid:

- Forgetting to add the carry from the previous column in column addition.
- Counting a carry when the column sum is exactly 9 (a carry needs a sum of 10 or more).
- Leaving out the divisors 1 and the number itself when listing divisors, or listing a divisor twice (as in $36 = 6 \times 6$).
- In Euclid's subtraction method, subtracting the larger number from the smaller or stopping before the two numbers are equal.
- Reading $m \bmod n$ as the quotient of $m \div n$; it is the remainder, and it is always less than n.
- Skipping steps when tracing an algorithm, so that a variable's value is not updated in the table.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **World of Algorithms** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 0	$65 = 5 \times 13 + 0$.
2	**(c)** 2	Common divisors: 1, 2.
3	**(c)** Al-Khwārizmī	Origin of the word.
4	**(d)** 56	Sum of even numbers up to 14.
5	**(c)** one of the numbers becomes 0	gcd(g, 0) = g.
6	**(b)** 1	Carries out of the columns: [0, 1].
7	**(c)** 620	Add column by column from the right, carrying 1 when a column sum exceeds 9.
8	**(a)** 2	Divisors of 23: 1, 23.
9	**(d)** 6	$90 = 12 \times 7 + 6$.
10	**(c)** 4	Common divisors: 1, 2, 4.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Check each j from 1 to 18: 18 numbers are checked; List $= [1, 2, 3, 6, 9, 18]$ (1); It has 6 entries (1).	1+1
12	$58 = 10 \times 5 + 8$, so remainder 8; $68 = 10 \times 6 + 8$, so remainder 8 (1); Adding the divisor does not change the remainder (1).	1+1
13	Add from the units column, carry 1 when a column sum exceeds 9; Columns with a carry: tens (1); $366 + 260 = 626$ (1).	1+1
14	Check each j from 1 to 23: 23 numbers are checked; List $= [1, 23]$ (1); It has 2 entries (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	divisors(10) $= [1, 2, 5, 10]$ (1); divisors(25) $= [1, 5, 25]$ (1); Common divisors $= [1, 5]$, greatest is 5 (1).	1+1+1
16	Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first (1); $(45, 30) \to (30, 15) \to (15, 15) \to (15, 0)$ (1); 3 subtractions, gcd $= 15$ (1).	1+1+1
17	Add from the units column, carry 1 when a column sum exceeds 9 (1); Columns with a carry: tens, hundreds, thousands (1); $6680 + 5984 = 12664$ (1).	1+1+1
	OR: divisors(27) $= [1, 3, 9, 27]$ (1); divisors(21) $= [1, 3, 7, 21]$ (1); Common divisors $= [1, 3]$, greatest is 3 (1).	1+1+1
18	Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first (1); $(45, 27) \to (27, 18) \to (18, 9) \to (9, 9) \to (9, 0)$ (1); 4 subtractions, gcd $= 9$ (1).	1+1+1
	OR: Add from the units column, carry 1 when a column sum exceeds 9 (1); Columns with a carry: tens, hundreds, thousands (1); $5856 + 6761 = 12617$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Even values: 2, 4, 6 (1); Total $= 2 + 4 + 6$ (2); Final total $= 12$ (2).	1+2+2
	OR: divisors(42) $= [1, 2, 3, 6, 7, 14, 21, 42]$ (1); divisors(48) $= [1, 2, 3, 4, 6, 8, 12, 16, 24, 48]$ (2); Common divisors $= [1, 2, 3, 6]$, greatest is 6 (2).	1+2+2
20	Even values: 2, 4, 6 (1); Total $= 2 + 4 + 6$ (2); Final total $= 12$ (2).	1+2+2
	OR: divisors(44) $= [1, 2, 4, 11, 22, 44]$ (1); divisors(50) $= [1, 2, 5, 10, 25, 50]$ (2); Common divisors $= [1, 2]$, greatest is 2 (2).	1+2+2

Mistakes to avoid:

- Forgetting to add the carry from the previous column in column addition.
- Counting a carry when the column sum is exactly 9 (a carry needs a sum of 10 or more).
- Leaving out the divisors 1 and the number itself when listing divisors, or listing a divisor twice (as in $36 = 6 \times 6$).

- In Euclid's subtraction method, subtracting the larger number from the smaller or stopping before the two numbers are equal.
- Reading $m \bmod n$ as the quotient of $m \div n$; it is the remainder, and it is always less than n.
- Skipping steps when tracing an algorithm, so that a variable's value is not updated in the table.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **World of Algorithms** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** greater than 9	Digit would not fit in one place.
2	**(c)** 20	Sum of even numbers up to 8.
3	**(c)** it is clear and unambiguous	Algorithms must be precise.
4	**(b)** 0	Carries out of the columns: [0, 0].
5	**(a)** 720	Add column by column from the right, carrying 1 when a column sum exceeds 9.
6	**(b)** 2	Divisors of 17: 1, 17.
7	**(c)** 7	$43 = 9 \times 4 + 7$.
8	**(d)** 2	Common divisors: 1, 2.
9	**(b)** $29 \bmod 6 = 5$	$29 = 6 \times 4 + 5$.
10	**(b)** 110	Sum of even numbers up to 20.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$73 = 5 \times 14 + 3$, so remainder 3; $78 = 5 \times 15 + 3$, so remainder 3 (1); Adding the divisor does not change the remainder (1).	1+1
12	Add from the units column, carry 1 when a column sum exceeds 9; Columns with a carry: units, tens (1); $446 + 397 = 843$ (1).	1+1
13	Check each j from 1 to 20: 20 numbers are checked; List $= [1, 2, 4, 5, 10, 20]$ (1); It has 6 entries (1).	1+1
14	$66 = 6 \times 11 + 0$, so remainder 0; $72 = 6 \times 12 + 0$, so remainder 0 (1); Adding the divisor does not change the remainder (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	divisors(38) $= [1, 2, 19, 38]$ (1); divisors(10) $= [1, 2, 5, 10]$ (1); Common divisors $= [1, 2]$, greatest is 2 (1).	1+1+1
16	Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first (1); $(32, 24) \to (24, 8) \to (16, 8) \to (8, 8) \to (8, 0)$ (1); 4 subtractions, gcd $= 8$ (1).	1+1+1
17	Add from the units column, carry 1 when a column sum exceeds 9 (1); Columns with a carry: thousands (1); $9110 + 2211 = 11321$ (1).	1+1+1
	OR: divisors(25) $= [1, 5, 25]$ (1); divisors(40) $= [1, 2, 4, 5, 8, 10, 20, 40]$ (1); Common divisors $= [1, 5]$, greatest is 5 (1).	1+1+1
18	Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first (1); $(52, 40) \to (40, 12) \to (28, 12) \to (16, 12) \to (12, 4) \to (8, 4) \to (4, 4) \to (4, 0)$ (1); 7 subtractions, gcd $= 4$ (1).	1+1+1
	OR: Add from the units column, carry 1 when a column sum exceeds 9 (1); Columns with a carry: hundreds (1); $3551 + 3723 = 7274$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Even values: 2, 4, 6, 8 (1); Total $= 2 + 4 + 6 + 8$ (2); Final total $= 20$ (2).	1+2+2
	OR: divisors(66) $= [1, 2, 3, 6, 11, 22, 33, 66]$ (1); divisors(40) $= [1, 2, 4, 5, 8, 10, 20, 40]$ (2); Common divisors $= [1, 2]$, greatest is 2 (2).	1+2+2
20	Even values: 2, 4, 6, 8, 10, 12 (1); Total $= 2 + 4 + 6 + 8 + 10 + 12$ (2); Final total $= 42$ (2).	1+2+2
	OR: divisors(50) $= [1, 2, 5, 10, 25, 50]$ (1); divisors(85) $= [1, 5, 17, 85]$ (2); Common divisors $= [1, 5]$, greatest is 5 (2).	1+2+2

Mistakes to avoid:

- Forgetting to add the carry from the previous column in column addition.
- Counting a carry when the column sum is exactly 9 (a carry needs a sum of 10 or more).
- Leaving out the divisors 1 and the number itself when listing divisors, or listing a divisor twice (as in $36 = 6 \times 6$).

- In Euclid's subtraction method, subtracting the larger number from the smaller or stopping before the two numbers are equal.
- Reading $m \bmod n$ as the quotient of $m \div n$; it is the remainder, and it is always less than n.
- Skipping steps when tracing an algorithm, so that a variable's value is not updated in the table.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND